Graphs of Linear Equations

Algebra · Edexcel IGCSE Mathematics A (4MA1) Paper 1 · Year 10 · worksheet

A+ Maths Hub

Name	Class	Date

Answer ALL questions.

Write your answers in the spaces provided. You must write down all the stages in your working.

Suggested time: 100 minutes · Total marks: 100

KEY RULES

Form	$y = mx + c$: m gradient, c y-intercept
Plotting	Make a table of values or use the intercepts
Rearranging	Write $ax + by = c$ as $y = mx + c$
Parallel / perpendicular	$m_1 = m_2$; $m_1 m_2 = -1$
Intersection	Solve the equations simultaneously

WORKED EXAMPLE 1

[2 marks]

Find the gradient and y-intercept of $3x + y = 6$.

Step 1 Rearrange to $y = mx + c$ $\quad y = -3x + 6$

Answer: **gradient -3, intercept $(0, 6)$**

WORKED EXAMPLE 2

[3 marks]

Find the equation of the line through $(0, 4)$ and $(2, 10)$.

Step 1 Gradient $\quad \dfrac{10 - 4}{2 - 0} = 3$

Step 2 y-intercept is 4 $\quad c = 4$

Answer: $y = 3x + 4$

WORKED EXAMPLE 3

[2 marks]

Does the point $(3, 13)$ lie on the line $y = 4x + 1$?

Step 1 Substitute $x = 3$ $\quad y = 4 \times 3 + 1 = 13$

Answer: **Yes,** $y = 13$

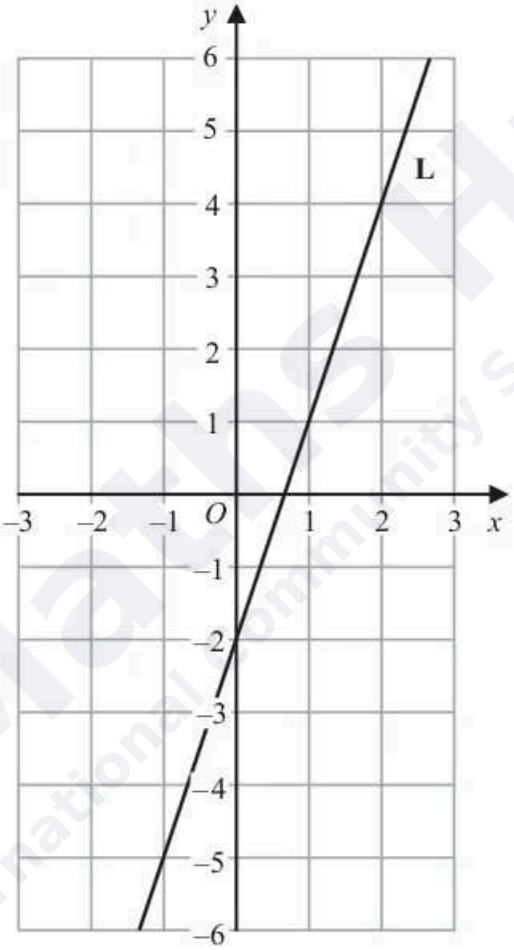
A+ Maths Hub International Community Schools

1 The point $(p, 50)$ lies on the line $y = 4x + 2$. Work out p.

(Total for question = 2 marks)

Practice question written for this worksheet (not from a past paper)

2 The line **L** is shown on the grid.



Find an equation for **L**.

Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · November 2020 · Question 6

3 A straight line passes through the points with coordinates (0, –3) and (2, 0)

Find an equation of the line.

Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · November 2023 · Question 9

4 A straight line, **L**, is parallel to the line with equation $y = 2 - 5x$
The line **L** passes through the point (0, 6)

Find an equation of the line **L**

Edexcel IGCSE Mathematics A (4MA1) · Paper 2HR · June 2025 · Question 10

5 Find the gradient of the straight line with equation $5x + 2y = 7$

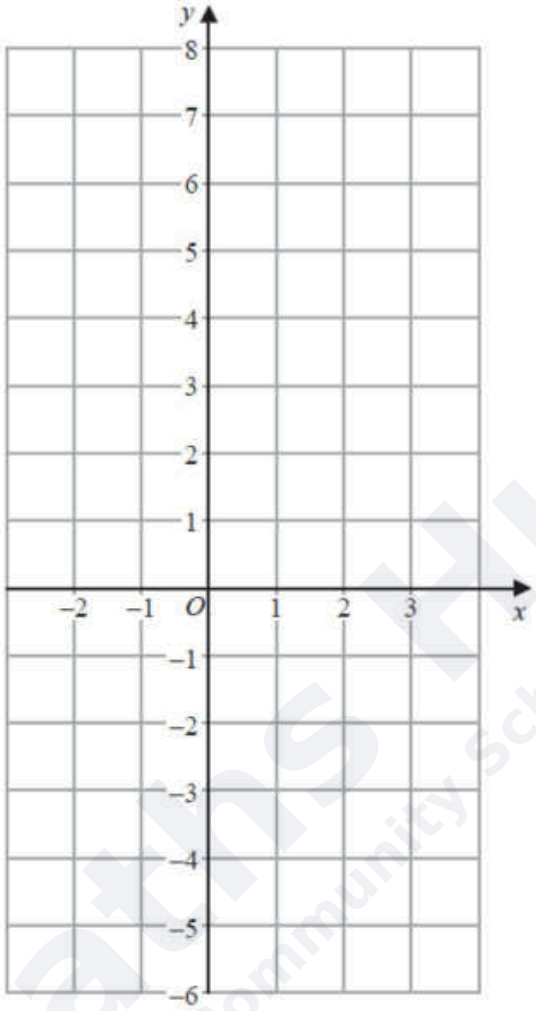
.....

Edexcel IGCSE Mathematics A (4MA1) · Paper 2HR · November 2020 · Question 12

A+ Maths Hub
International Community Schools

Graphs of Linear Equations

6 On the grid below, draw the graph of $y = 1 - 2x$ for values of x from –2 to 3

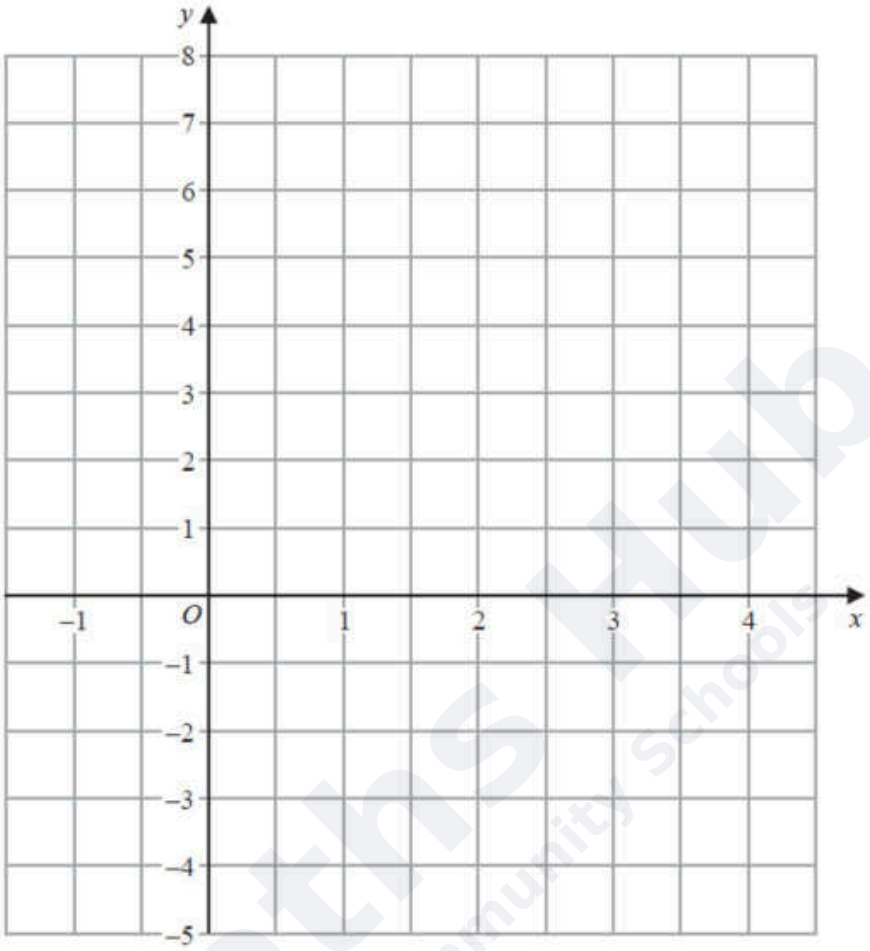


(Total for question = 3 marks)

Edexcel IGCSE Mathematics A (4MA1) · Paper 1F · January 2023 · Question 11

A+ Maths Hub International Community Schools

7 On the grid, draw the graph of $y = 4 - 2x$ for values of x from -1 to 4

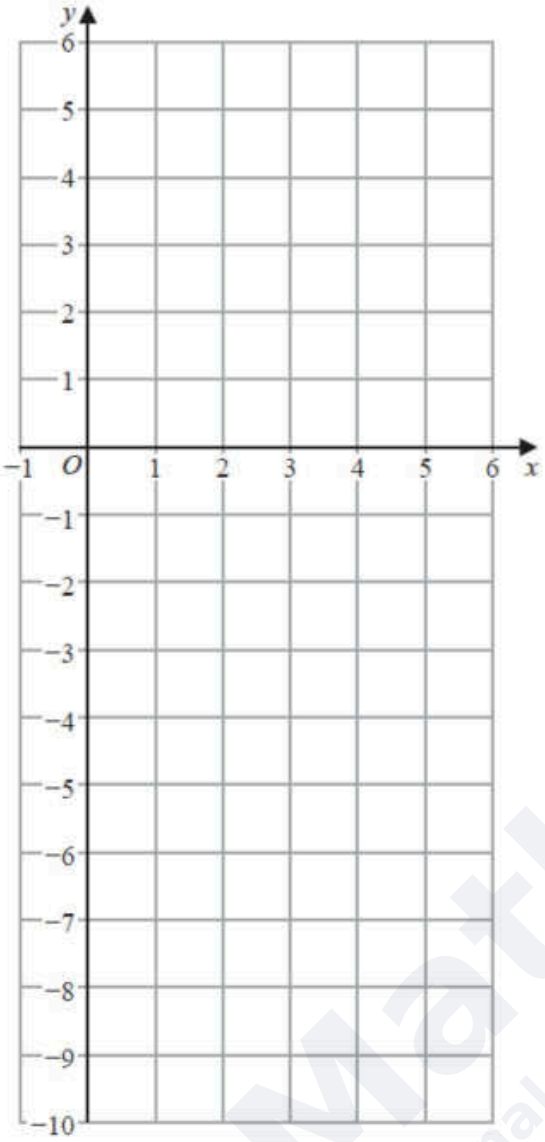


(Total for question = 3 marks)

Edexcel IGCSE Mathematics A (4MA1) · Paper 1FR · June 2023 · Question 12

A+ Maths Hub International Community Schools

8 On the grid, draw the graph of $y = -2x + 3$ for values of x from –1 to 5

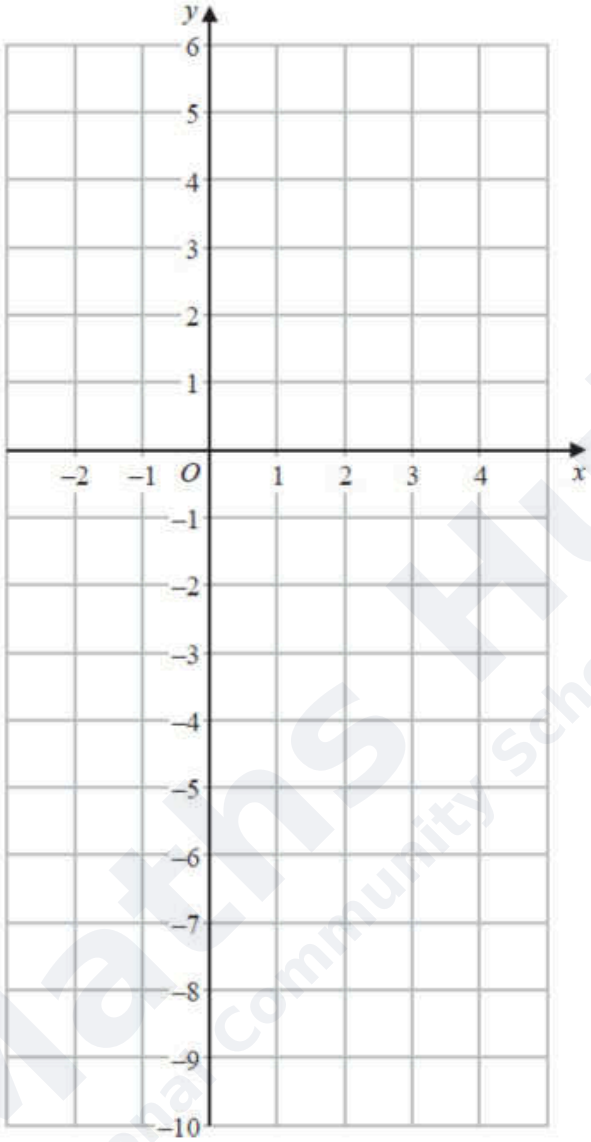


(Total for question = 3 marks)

Edexcel IGCSE Mathematics A (4MA1) · Paper 1F · June 2022 · Question 13

A+ Maths Hub
International Community Schools

9 On the grid, draw the graph of $y = 2x - 3$ for values of x from –2 to 4

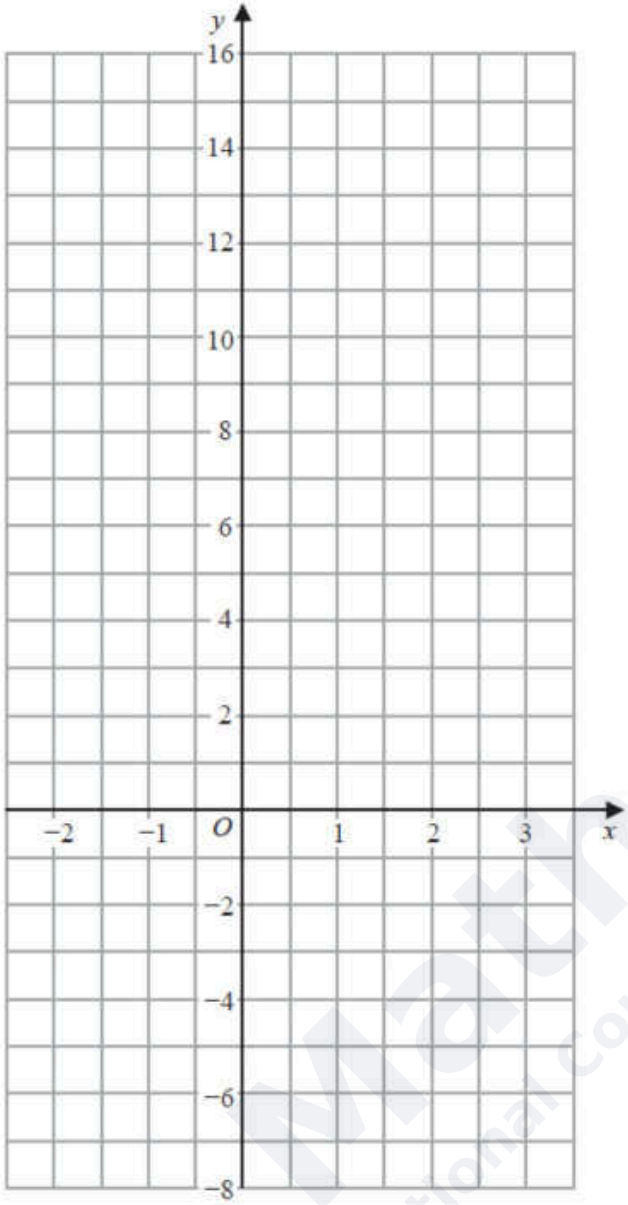


(Total for question = 3 marks)

Edexcel IGCSE Mathematics A (4MA1) · Paper 1F · June 2024 · Question 13

A+ Maths Hub International Community Schools

10 On the grid, draw the graph of $y = 7 - 4x$ for values of x from -2 to 3

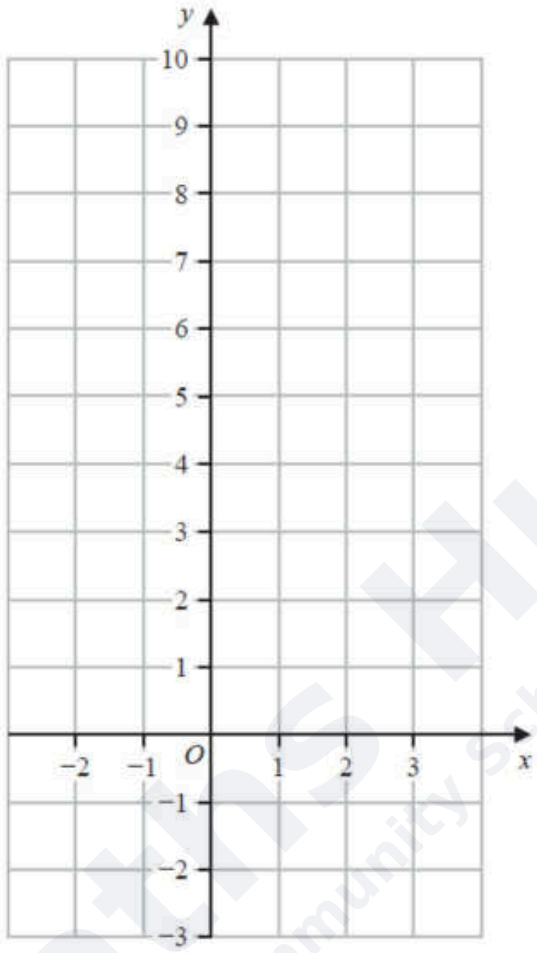


(Total for question = 3 marks)

Edexcel IGCSE Mathematics A (4MA1) · Paper 1F · November 2020 · Question 16

A+ Maths Hub International Community Schools

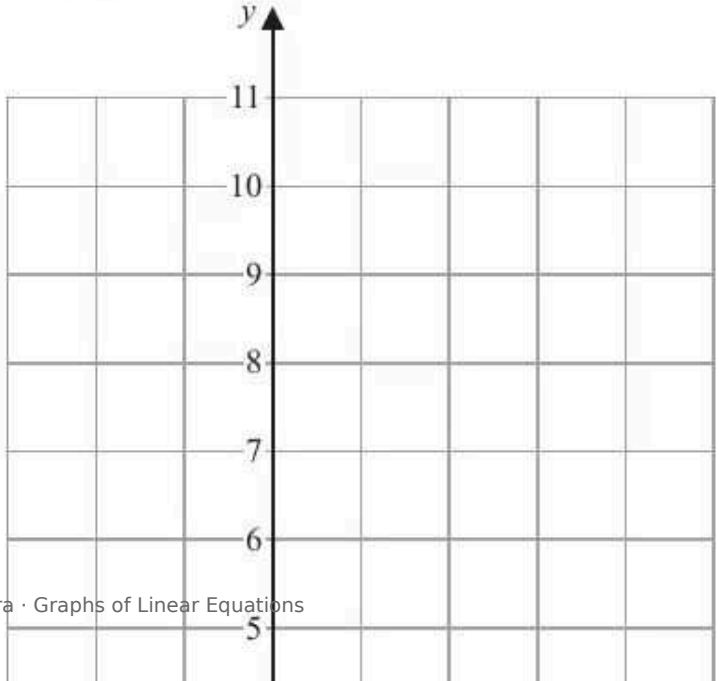
11 On the grid, draw the graph of $y = 5 - 2x$ for values of x from -2 to 3



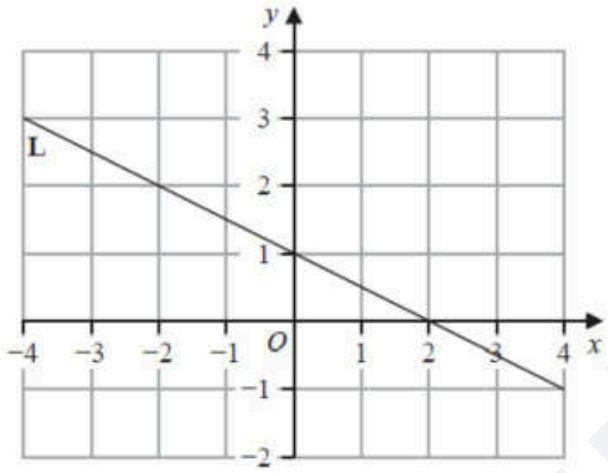
(Total for question = 3 marks)

Edexcel IGCSE Mathematics A (4MA1) · Paper 1F · June 2026 · Question 16

12 On the grid, draw the graph of $5x + 2y = 10$ for values of x from -2 to 4



13 Line **L** is drawn on the grid.



Find an equation for **L**
Give your answer in the form $y = mx + c$

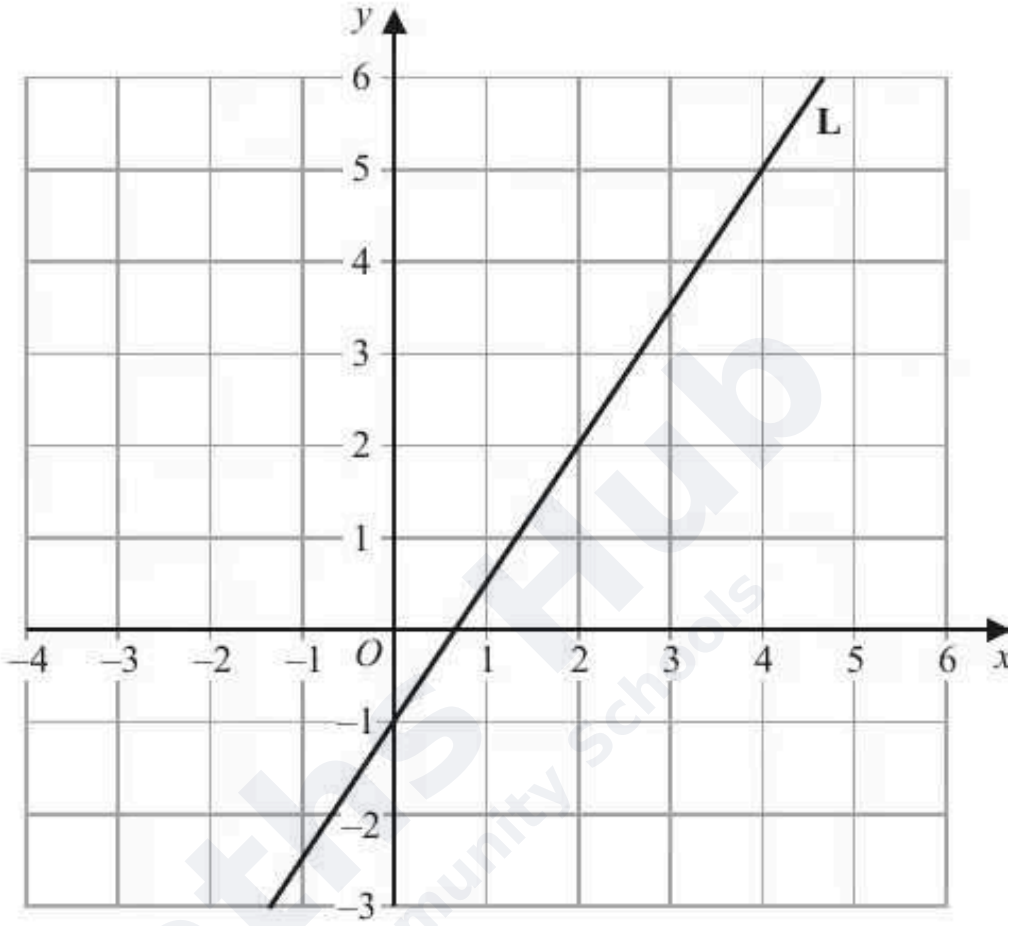
.....

(Total for question = 3 marks)

Edexcel IGCSE Mathematics A (4MA1) · Paper 1HR · June 2024 · Question 9

A+ Maths Hub
International Community Schools

14 Line **L** is drawn on the grid.

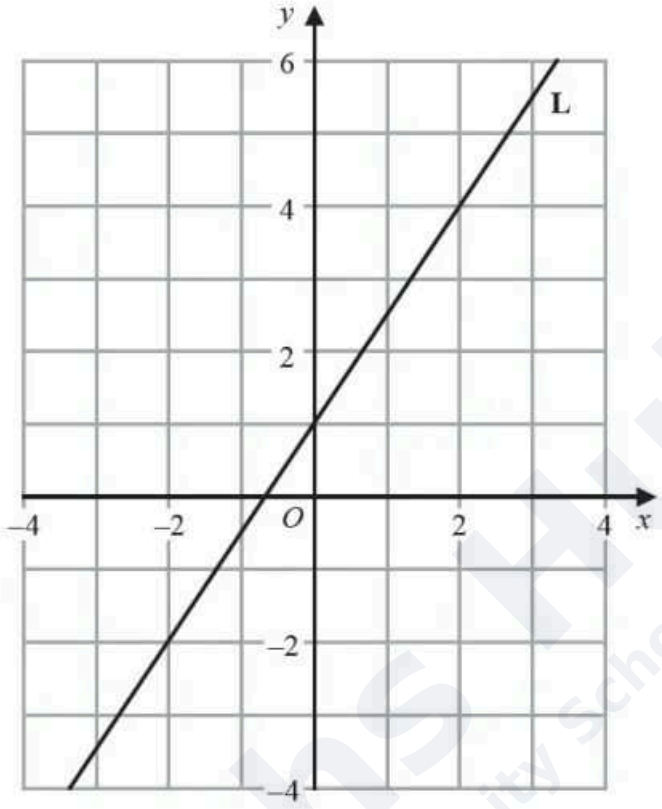


Find an equation for **L**
Give your answer in the form $y = mx + c$

Edexcel IGCSE Mathematics A (4MA1) · Paper 1HR · June 2022 · Question 10

A+ Maths Hub International Community Schools

15 The line **L** is drawn on the grid.

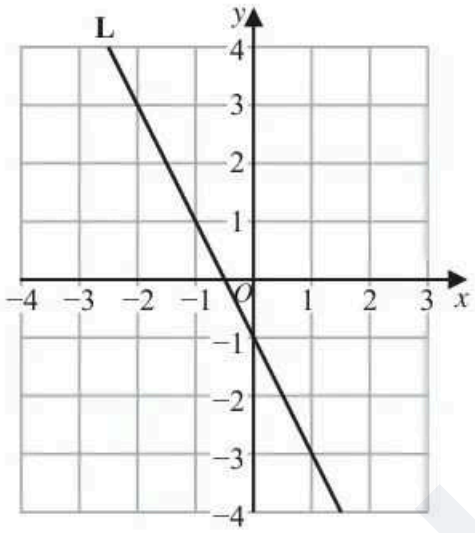


Find an equation for **L**.

A+ Maths Hub International Community Schools

Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · January 2019 · Question 10

16 Line **L** is drawn on the grid.



Find an equation for **L**.

Edexcel IGCSE Mathematics A (4MA1) · Paper 2HR · June 2019 · Question 11

17 Find an equation of the line that is parallel to the line $y = 4 - 2x$ and passes through the point (3, 7)

Edexcel IGCSE Mathematics A (4MA1) · Paper 4H · January 2016 · Question 12

A* Maths Hub
International Community Schools

18 The straight line **L** has equation $2y + 7x = 10$

(a) Find the gradient of **L**

.....

(2)

(b) Find the coordinates of the point where **L** crosses the y-axis.

(..... , ...)

(1)

(Total for question = 3 marks)

Edexcel IGCSE Mathematics A (4MA1) · Paper 1F · January 2022 · Question 25

19 Point A has coordinates (5, 8)
Point B has coordinates (9, –4)

(a) Work out the gradient of AB.

.....

(2)

The straight line **L** has equation $y = -4x + 5$

(b) Write down the gradient of a straight line that is perpendicular to **L**.

.....

(1)

(Total for question = 3 marks)

Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · January 2021 · Question 13

A+ Maths Hub International Community Schools

20 A is the point with coordinates $(-5, 12)$

B is the point with coordinates $(19, -48)$

Find an equation of the straight line that passes through the points A and B

Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · June 2023 · Question 13

21 The straight line **L** has equation $y = 4x + 7$

The straight line **M** is perpendicular to **L** and passes through the point with coordinates $(8, 1)$

Find an equation for **M**
Give your answer in the form $y = mx + c$

Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · November 2025 · Question 20

A+ Maths Hub
International Community Schools

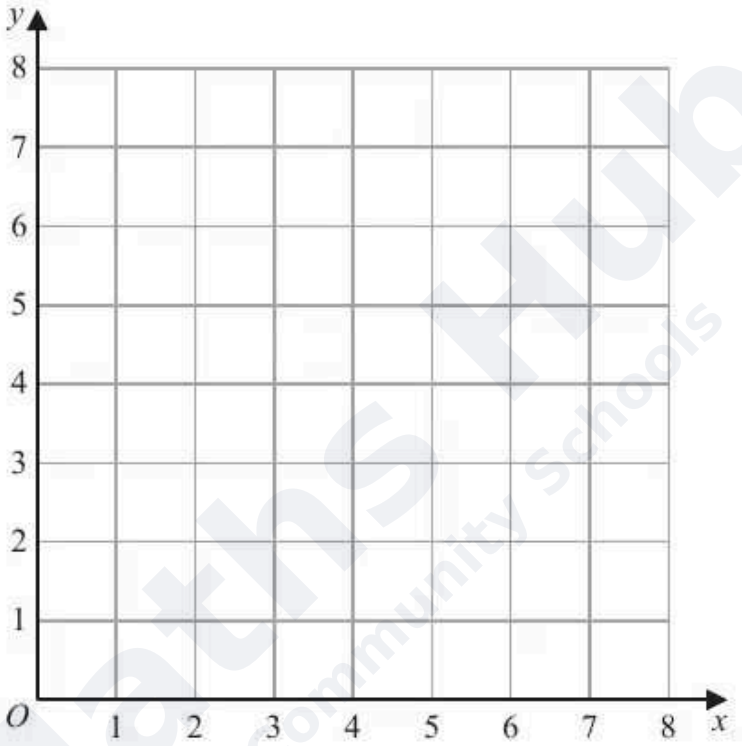
22 (a) On the grid, draw the straight line with equation

(i) $y = 2$

(ii) $x = 6$

(iii) $y = x + 1$

Label each line with its equation.



(3)

(b) Show, by shading on the grid, the region that satisfies all three of the inequalities

$y \geqslant 2$ $x \leqslant 6$ $y \leqslant x + 1$

Label the region **R**

(1)

(Total for Question 2 is 4 marks)

Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · June 2024 · Question 2

A+ Maths Hub International community Schools

23 (a) Find the gradient of the line with equation $3y - 2x = 6$

.....

(2)

(b) Find an equation of the line with gradient -3 that passes through the point $(2, 5)$.

.....

(2)

Edexcel IGCSE Mathematics A (4MA1) · Paper 3H · January 2015 · Question 10

24 The straight line **L** passes through the points $(-2, 3)$ and $(6, 9)$

Find an equation of the line that is parallel to **L** and passes through the point $(5, -1)$
Give your answer in the form $ax + by = c$ where a, b and c are integers.

.....

Edexcel IGCSE Mathematics A (4MA1) · Paper 3H · June 2016 · Question 13

A+ Maths Hub International Community Schools

CHALLENGE QUESTIONS

25 The straight line **L** has equation $3x - 2y = 15$

(a) Find the gradient of **L**.

.....

(3)

(b) Find the coordinates of the point where **L** crosses the y-axis.

(..... , ...)

(1)

(c) Find an equation of the line that is parallel to **L** and crosses the x-axis at $(-2, 0)$

.....

(2)

Edexcel IGCSE Mathematics A (4MA1) · Paper 4H · January 2017 · Question 12

A+ Maths Hub
International Community Schools

26 *PQRS* is a square.
PR is a diagonal of the square.

P is the point with coordinates (4, 7)
R is the point with coordinates (8, −5)

Find an equation of the straight line that passes through the points *Q* and *S*
Give your answer in the form $ay = bx + c$ where *a*, *b* and *c* are integers.

.....

(Total for question = 5 marks)

Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · June 2025 · Question 25

27 Triangle *HJK* is isosceles with *HJ* = *HK* and $JK = \sqrt{80}$

H is the point with coordinates (−4, 1)
J is the point with coordinates (*j*, 15) where $j < 0$
K is the point with coordinates (6, *k*)

M is the midpoint of *JK*.
The gradient of *HM* is 2

Find the value of *j* and the value of *k*.

j = ...

k = ...

(Total for question = 6 marks)

Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · January 2020 · Question 22

A+ Maths Hub International Community Schools

28 $ABCD$ is a kite.

$AB = AD$ and $CB = CD$

The point B has coordinates $(k, 1)$ where k is a negative constant.
The point D has coordinates $(8, 7)$

The straight line **L** passes through the points B and D
The straight line **L** is parallel to the line with equation $5y - 3x = 6$

Find an equation of AC
Give your answer in the form $px + qy = r$ where p, q and r are integers.
Show your working clearly.

.....

(Total for question = 6 marks)

Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · January 2023 · Question 23

A+ Maths Hub
International Community Schools

29 P and Q are two points.

The coordinates of P are $(-1, 6)$

The coordinates of Q are $(5, -4)$

Find an equation of the perpendicular bisector of PQ.
Give your answer in the form $ax + by + c = 0$ where a, b and c are integers.

Edexcel IGCSE Mathematics A (4MA1) · Paper 2HR · November 2020 · Question 23

TOTAL FOR PAPER IS 100 MARKS

A+ Maths Hub International Community Schools